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USN

R. V. COLLEGE OF ENGINEERING

Autonomous Institution affiliated to VTU

III Semester B. E. Examinations Nov/Dec-17

Electronics and Communication Engineering

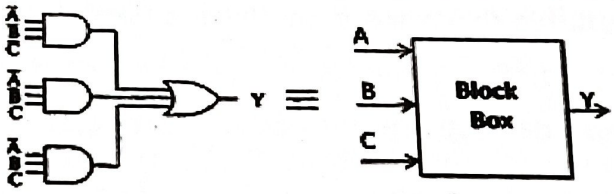
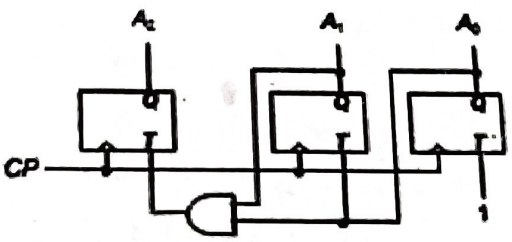
ANALYSIS AND DESIGN OF DIGITAL CIRCUITS

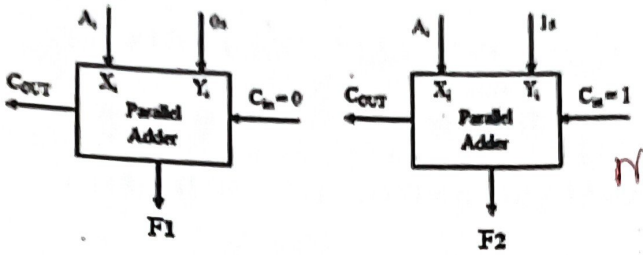
Time: 03 Hours *Maximum Marks: 100*

Instructions to candidates:

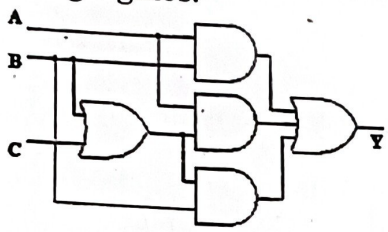
1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B.In Part B question number 2, 7 and 8 are compulsory. Answer any one full question from 3 and 4 & one full question from 5 and 6

PART-A

1	1.1	The figure of merit for a digital circuit is the product of __power__ and __Delay__.	01
	1.2	In a 5 variable K-map, cell 3 is adjacent to __1__, __2__, __7__, __11__ and __19__.	01
	1.3	Explain the following terms with suitable examples: i) Redundant group ii) Prime implicants.	02
	1.4	Fill up the black box in fig 1.4, with minimum number of logic gates.  $Y = \bar{A}\bar{B} + \bar{A}C$ Fig 1.4	02
	1.5	Propose the controlled inverter for 8-bit adder/subtractor circuit.	02
	1.6	Write the characteristic equation for a $JK - FF$. $J\bar{Q} + \bar{K}Q$	01
	1.7	Write the excitation table for a $T - FF$. 0, 1, 1, 0	01
	1.8	Show the arrangement for a $Mod - 15$ counter, using a 4 bit counter.	02
	1.9	Modulus of acounter means _____.	01
	1.10	A sequential circuit with $3 - FFs$ will have __2^3__ number of possible states.	01
	1.11	Prepare the function table for the digital circuit shown in fig 1.11.  000 001 010 011 100 101 110 111 Fig 1.11	02
	1.12	In an ALU function select control lines distinguish between __Arithmetic__ and __Logic Unit__.	01

	1.13	Implement the overflow flag for an 8- bit *ALU*.	01
	1.14	Check for the equivalence of the functions F_1 and F_2 in fig 1.14. identify the functionality.  Transfer A Fig 1.14	02

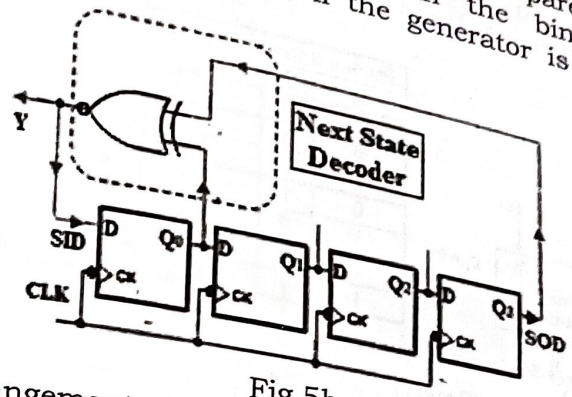
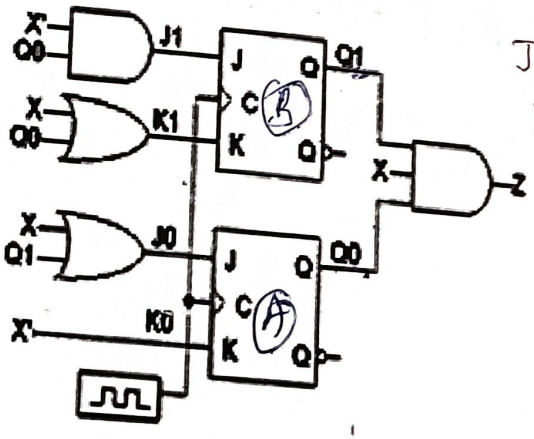
PART-B

2	a	Simplify the logic circuit shown in fig 2a and implement the simplified logic circuit using basic logic gates.  $Y = B + AC$ Fig 2 a	06
	b	Simplify the following Boolean function using K-map and obtain the most optimal sum of products (*SOP*) expression. $f(a,b,c,d,e) = \Sigma m(0,1,4,5,6,7,9,12,13,14,15,16,17,20,21,22,23,25,29)$ $\bar{b}c + \bar{b}\bar{d} + \bar{d}e + \bar{a}c$	06
	c	Using suitable diagrams demonstrate various voltage levels for a *TTL* gate.	04
3	a	Design and develop a 4 bit adder/subtractor using a parallel adder. The circuit has a control input S that controls addition or subtraction operation, $S = 0$ results in addition and $S = 1$ results subtraction operation.	06
	b	Implement the Boolean function $F(A\,B\,C\,D) = \Sigma m(1,2,3,4,11,12,13,14,15)$, using 8×1 multiplier. Show the design procedure. D, 1, $\bar{D}$, 0, 0, D, 1, 1	06
	c	Develop 2×2 array multiplier using *AND* gates and half adders. Explain the concept with an example. 4 AND gates & 2 HAs	04
		OR	
4	a	Design and develop a 4-bit carry look ahead adder. Comment on the circuit propagation delay.	06
	b	Realize a full adder using a 3:8 decoder and comment on the implementation.	06
	c	Develop the truth table for a 2 bit binary magnitude comparator and obtain the expression for the outputs.	04
5	a	What is a race around condition? Explain with the help of timing wave forms. What are the possible ways to eliminate this problem? Explain.	06

$G_i = a$
$P_i =$
$C_i =$

$A<B = \cancel{B_0 A_1 \bar{B}_1 \bar{A}_1 A_0} + \bar{A}_1 \bar{A}_0 B_0 + \bar{A}_1 B_1 + \bar{A}_0 B_1 B_0$

$A>B = A_1 \bar{B}_1 + A_0 \bar{B}_1 \bar{B}_0 + \bar{B}_0 A_1 A_0$

		For the sequence generator shown in fig 5b, prepare the function table, and estimate the randomness in the binary sequence generated. Discuss the situation, if the generator is initiated with value 0000.  $y = \overline{Q_0 \oplus Q_3}$ Fig 5b	
	c	Draw the arrangement for a serial adder and comment on the hardware.	06
		OR	
6	a	With a neat logic diagram and function table, explain the working of a *JK* Flip-Flop. The arrangement should include preset and clear facilities.	04
	b	Design a sequence generator to generate the binary sequence 1101011.	08
7	a	Obtain the governing equations and the state for the sequential circuit given in figure 7a,. Briefly explain the same.  $J_1 = \overline{X} Q_0$ $J_0 = X + Q_1$ $K_1 = X + Q_0$ $K_0 = \overline{X}$ Fig 7a	08
	b	Design a counter using *JK FFs* to count the non-binary sequence and briefly explain: 000 → 001 → 010 → 100 → 101 → 110 → 000	08
8	a	Implement the models required to design an ALU. i) Status register ii) Combinational shifter	08

7b) $J_A = B$ $K_A = B$

$J_B = C$ $K_B = 1$

$J_C = \overline{B}$ $K_C = 1$

$\overline{A}$	0	1	1	1	1	0	0	0
A	0	0	0	1	1	1	1	1
	Q_0	Q_1	Q_2	Q_3	Q_4	Q_5	Q_6	Q_7
	0	$\overline{A}$	$\overline{A}$	1	1	A	A	A

b	The arrangement shown in fig 8b depicts the transfer operation. Develop and demonstrate similar arrangements to realize the set of arithmetic operations listed in the table 8b.	08

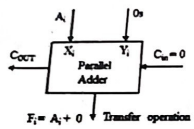


Fig. 8b: Transfer A (Row-1)

Function Select		Input Carry	Input	Output	Function realized
S_1	S_0	C_{in}	Y_i	F_i	
0	0	0	0	$F_i = A_i$	Transfer A
0	0	1	0	$F_i = A_i+1$	Increment A
0	1	0	B_i	$F_i = A_i+B_i$	Add B to A
0	1	1	B_i	$F_i = A_i+B_i+1$	Add B to A with carry
1	0	0	$\overline{B}_i$	$F_i = A_i+\overline{B}_i$	1s complement addition
1	0	1	$\overline{B}_i$	$F_i = A_i+\overline{B}_i+1$	2s complement addition
1	1	0	All 1s	$F_i = A_i-1$	Decrement A
1	1	1	All 1s	$F_i = A_i$	Transfer A

Table. 8b: Arithmetic operations for ALU design

$A = B \quad (\overline{A_0 \oplus B_0}) + (\overline{A_1 \oplus B_1})$

$A < B \quad \overline{A_1} B_1 + \overline{A_1}\,\overline{A_0} B_0 + \overline{A_0} B_1 B_0$

$A > B \quad A_1 \overline{B_1} + A_0 \overline{B_1}\,\overline{B_0} + A_1 A_0 \overline{B_0}$